

Homework 2 — How Machines Learn

Bong Study Hub — Foundation Batch 2026 · Class 02: How Machines Learn

Estimated time: 30–45 minutes **Tools needed:** pen and paper, or a simple document — nothing else

Prerequisite: Class 02. No coding, no software, no advanced AI knowledge required.

What This Assignment Is About

In class, we answered one question:

"How can a machine make a decision when nobody explicitly wrote the rule for every situation?"

We saw that writing rules by hand works well when a human can clearly describe the logic — but becomes difficult once the number of situations and exceptions grows. We saw an alternative:

DATA → LEARNING → MODEL → PREDICTION

— give the system examples instead of rules, and let it find useful patterns on its own. We reasoned through this with a spam filter and with the cats-vs-dogs activity, and we saw that a prediction is a best guess, not a guarantee — and that the data behind it matters.

This assignment asks you to **explain these ideas in your own words** and apply them to something new. There is no coding, no math, and no single "correct sentence" to memorize — if your reasoning is sound, your answer is good, even if it doesn't use the exact words from class.

Instructions

- Complete this **independently**, in **your own words**.
- **No coding is required**, and none should appear in your answers.
- **No advanced AI knowledge is required** beyond what Class 02 covered.
- You may refer to your **Student Notes** while answering.
- **Estimated time:** 30–45 minutes.
- If you're unsure about an answer, **explain your reasoning anyway** rather than leaving it blank — partial thinking is worth far more to your instructor (and to you) than a blank space.
- Use AI tools only to check your understanding *after* you've tried answering yourself — not to generate your answers. This assignment exists to check *your* thinking, not a tool's.

Question 1 — Rules (2 points)

Imagine you are building a spam filter using only human-written rules.

Give two rules you might write to identify spam, and give one example where each rule could fail.

Question 2 — Why Rules Become Difficult (2 points)

Why can writing every rule by hand become difficult for some real-world problems?

Explain the idea of growing situations and exceptions in your own words — you don't need to use any particular phrase from class.

Question 3 — The Pivot (1 point)

If writing every rule becomes impractical, what could we give the computer instead?

Question 4 — Core Pipeline (3 points)

Here are the four stages of Class 02's core pipeline, scrambled:

PREDICTION

MODEL

DATA

LEARNING

- Put them in the correct order.
- Explain each stage in one short sentence.

Question 5 — Traditional Programming vs. Machine Learning (2 points)

Fill in the blanks:

Traditional Programming:

INPUT → _____ → OUTPUT

Machine Learning:

_____ → LEARNING → MODEL → _____

Then, **in one or two sentences, what is the main difference between the two approaches?**

Question 6 — Cats vs. Dogs (2 points)

Suppose a system has seen many labeled examples of cats and dogs. A new image arrives.

Explain, in simple language, what the system would do to decide whether the new image is a cat or a dog.

Question 7 — Prediction ≠ Certainty (2 points)

A spam filter marks a genuine email as spam.

Does this mean the system was not using Machine Learning?

Why could this happen?

Question 8 — Data Quality (2 points)

Suppose a cat/dog system sees only photographs of **small white cats** and **large black dogs** during learning.

What might happen when it sees a large white cat? Explain why.

Question 9 — Misconception Correction (2 points)

For each statement below, say whether it is **correct** or **incorrect**. If it is incorrect, **rewrite it in a better way**.

- A. "Machine Learning replaces programming."
- B. "If a model makes a prediction, the prediction must be correct."
- C. "Machine Learning means the computer thinks like a human."

Question 10 — Apply It to an Everyday Example (2 points)

Choose **ONE** everyday system from this list:

- Spam detection
- Recommendation systems
- Maps / navigation
- Face unlock
- Generative AI / ChatGPT

You are **not** being asked to explain how the real product actually works internally — just to apply the simple Class 02 mental model to it, conceptually.

Chosen system: _____

1. What kind of input/examples might the system use?
2. What useful patterns might matter?
3. What kind of prediction/output could it produce?
4. Why might the prediction still be wrong?

Optional Reflection — Not Scored

This section is entirely optional and will not affect your grade.

Which idea from today's class changed the way you think about AI? Explain in 2–3 sentences.

Submission Format

Choose whichever is easiest for you:

- **Option A:** Handwrite your answers, then photograph or scan them.
- **Option B:** Type your answers in a simple document (Word, Google Docs, plain text).
- **Option C:** Submit as a simple PDF.

No special formatting tools, GitHub, or programming software required. Please label each answer with its question number.

Before You Submit — Self-Check

- ☐ I answered all 10 questions in my own words.
- ☐ I did not leave any answer blank — where unsure, I explained my reasoning instead.
- ☐ I did not use code or programming syntax anywhere.
- ☐ I did not rely on an AI tool to generate my answers for me.
- ☐ I reviewed my work before submitting.